Structure-Guaranteed SAR Imagery via Spatially-Variant Morphology Regularization in ADMM Manner

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***Abstract*— Conventional sparsity-driven synthetic aperture radar (SAR) imagery proceeds via ℓ_1 regularization, or named by least absolute shrinkage and selection operator (LASSO). However, followed by the enhanced sparse feature, structures of the scenes or targets of interest in weak scattering are easily lost. Therefore, it becomes difficult to make use of the high-resolution SAR data, and even high costs have been paid for the resolution. In this article, a novel structure-guaranteed SAR (SG-SAR) imaging algorithm is proposed by utilizing the morphology metric for the cluster feature of the scatterers of the scenes/targets of interest. By introducing structural priors in terms of morphology norm, the intended structure features can be highlighted via convex regularization. More specifically, to accommodate complicated scenarios or targets, the structure element (SE) in the morphology regularizer is designed to be spatially variant under structure tensor representation. Different from conventional convex optimizations, the proposed SG-SAR algorithm is solved under the alternating direction method of multipliers (ADMMs), which is flexible to incorporate with the super-resolution imagery. In such cases, both sparse and structural features can be simultaneously enhanced, even with limited measurements and in a low signal-to-noise ratio (SNR). Superior convergence and robustness can be guaranteed. Moreover, a grouping mask scheme is used to accommodate the complex-valued SAR data. Finally, both simulated and measured SAR data are applied for the validation. Comparisons with the conventions are performed in terms of phase transition analysis, so as to verify the superiority of the proposed algorithm both qualitatively and quantitatively.**

***Index Terms*— Alternating direction method of multipliers (ADMMs), morphology regularization (MR), proximity operator, structure-guaranteed synthetic aperture radar (SG-SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a microwave imaging tool that is capable of working in all weather and all day-night [1]. The range resolution of SAR depends on the bandwidth of the transmitting pulses, and the azimuthal resolution relies on the size of the synthetic aperture. The synthetic aperture of SAR is generated by the cooperative movement between the radar and the target of interest [2]. To achieve elaborated structure features of the observing scenes or targets, the improvement of the resolution of the SAR system would be the most effective way. The higher the SAR resolution, the finer the target features can be explored [3], [4].

However, in order to improve the resolution, the wider bandwidth of transmitted pulses, and Doppler frequency, more sophisticated auxiliaries are required. It will definitely lead to a high cost of the system and a large scale of complexity. Furthermore, the algorithms of SAR imagery are conventionally developed in a Fourier transformation (FT) manner, which limits detailed features of the scenes or targets to be distinguished [5]. With the emergence of compressive sensing (CS) theory, dominant features in sparsity can be highlighted during the sparse SAR imagery recovery [6], [7], [8]. Commonly, by imposing the sparse prior in ℓ_1 norm, the sparse recovery problem is formulated in the least absolute shrinkage and selection operator (LASSO) model [9]. The solutions for the LASSO model mainly include a greedy manner and convex optimization. Although the greedy manner is efficient, it is of low accuracy [10], [11], [12]. The convex optimization is with superior convergence, but the computational efficiency is difficult to be balanced with the accuracy [13], [14], [15]. Although the LASSO model is popular in sparsity-driven SAR imagery, the elaborated features in weak scattering are easily lost when dominant sparse scatters are enhanced. Therefore, further target recognition will be influenced due to the insufficient recovery of the target features.

Refined features can be known as the clustering features of scatters, such as structure, shape, contour, and details about the features of the interesting targets. To explore refined features, the LASSO model has been developed into a series of extended versions. By replacing the sparsity of ℓ_1 norm regularizer with a grouped penalty in the ℓ_1/ℓ_2 mixed regularizer, the group LASSO model [16] is established. Although the group LASSO is capable of highlighting the clustering features, it cannot adjust overlapped structures. To this end, an overlapped group LASSO (OGL) is proposed in [17], which is capable of enhancing the sparsity between the groups and smoothing inner groups. However, it is inflexible to accommodate complicated structures in practice [18]. Friedman *et al.* [19] proposed a sparse group LASSO (SGL) to achieve the sparsity inside and outside the groups, simultaneously. Considering the

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complexity of the rich content in real scenarios, the structures would be both temporally and spatially variant. Therefore, both OGL and SGL models would be inflexible to adjust to the complicated features to be enhanced.

Another class of methods to achieve structural recovery is known as statistical Bayesian inference [20], [21], [22], [23]. By imposing the prior in a statistical manner, the posterior of the feature can be derived under Bayesian inference. When Laplacian distribution is encoded as a prior, the sparse feature can be enhanced under the hierarchical Bayesian model. The algorithm is known as the sparse Bayesian learning (SBL) or Bayes CS (BCS) [24]. Regarding the structural feature enhancement, a generalized pattern-coupled Bayesian model is proposed in [20]. It is flexible to capture underlying block-sparse structures without requirements of prior knowledge of the block partition. In [21], a spatially correlated prior is proposed to encourage the continuity structure of the target. In such a case, structural feature enhancement under the statistical Bayesian framework has received wide attention. However, the solvability of the posterior becomes an issue when encoding complicated priors. Besides, the computational load is high, which limits practical applications. Therefore, it is valuable to introduce novel models to formulate structural features to be explored.

MM is a powerful tool to capture structural details of the target to be imaged [25], [26]. It employs the structure element (SE) to acquire the structural features [27], [28]. The SE is applied as a probe to traverse the imaging scene/target. Besides, the interrelationships between the clusters can be clarified. MM provides multiple morphological operators to capture complicated features, and it is powerful to formulate refined structure priors [29]. However, the conventional MM typically follows a fixed SE, which may lead to low flexibility. Moreover, the conventional solution for the MM regularization is inflexible to incorporate multiple features to be enhanced. Low computation efficiency and poor convergence performance limit the practicability of the conventional MM regularization.

In this article, a novel structure-guaranteed SAR (SG-SAR) imaging algorithm is proposed based on spatially variant morphology regularization. By introducing a structural prior in terms of spatially variant morphology norm, refined features can be highlighted even in complicated structures. Specifically, the structure tensor is applied to construct the intended spatially variant SE. In such a case, the varying features can be captured by the proposed SE in an auto-fitting manner. In order to recover the structural features while retaining super-resolution capability, the proposed SG-SAR algorithm operates in the alternating direction method of multipliers (ADMMs) framework. The advantage of the ADMM framework is that it can coordinate two regularizers in spatially variant morphology and ℓ_1 norms. The solution procedure is designed in a "distributed-cooperative" manner, so that multiple regularizers can be optimized and balanced properly. Since the regularizations are surrogated into convex optimizations, an optimal solution can be guaranteed. In addition, a grouping mask scheme is used to accommodate the complex-valued SAR data. Finally, both simulated and raw SAR data are applied to verify the proposed algorithm, and comparisons with the conventions are performed to demonstrate the superiority of the proposed algorithm both quantitatively and qualitatively.

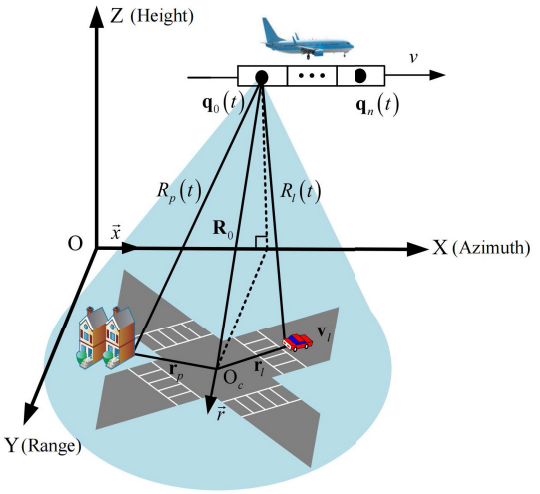


Fig. 1. Generalized airborne SAR geometry.

The rest of this article is organized as follows. Section II introduces SAR and ground moving target imaging (GMTIm) geometry, and a generalized signal model is given. Section III briefly reviews conventional sparsity-driven algorithms, structure recovery algorithms, generalized ADMM framework, and MM theory. The intended SG-SAR algorithm is proposed in Section IV. The processing flowchart is shown and explained in Section V. Experiments are performed and evaluated in Section VI. Finally, conclusion are drawn in Section VII.

II. Generalized SAR Geometry and Signal Model

Fig. 1 depicts a generalized airborne SAR geometry, where the Cartesian coordinates XYZ-O are established with the point O as the origin of the coordinates. Assume that the airborne SAR platform flies in the direction of the X-axis (azimuth) with the constant velocity v. During the data acquisition, the radar works in the spotlight mode [30]. In this article, to accommodate moving target imagery, a multichannel SAR antenna is employed. In such cases, both the echoes from the stationary scene and moving targets are collected. Suppose $\mathbf{q}_0(t)$ is the displacement vector of the reference SAR channel and $\mathbf{q}_n(t)$ is that of the nth SAR channel, where $\mathbf{q}_0(t) = (vt)\vec{\mathbf{x}}$, $\mathbf{q}_n(t) = \mathbf{q}_0(t) + (nd)\vec{\mathbf{x}}$, and $\vec{\mathbf{x}}$ is the unit vector in the azimuthal direction.

Assume the nearest slant range from the flight track to the scene center $\mathbf{O}_c$ is $\mathbf{R}_0 = R_0\vec{\mathbf{r}}$, where $\vec{\mathbf{r}}$ is the unit vector in the slant-range direction. The slant-range history from the radar to the pth stationary scatter can be given as

$$\begin{aligned} R_p(t) &= \left|\mathbf{R}_0 + \mathbf{r}_p - \mathbf{q}_0(t)\right| \\ &= |\mathbf{R}_0 - \mathbf{q}_0(t)| + \frac{\mathbf{R}_0 - \mathbf{q}_0(t)}{|\mathbf{R}_0 - \mathbf{q}_0(t)|}\mathbf{r}_p + O(\mathbf{r}_p) \end{aligned} \quad (1)$$

where $|\cdot|$ is the modulus operator of a vector, $\mathbf{r}_p = r_p^r\vec{\mathbf{r}} + r_p^x\vec{\mathbf{x}}$ is the offset vector of the stationary scatter from the scene center, and r_p^x and r_p^r are azimuthal and slant-range positions

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of the stationary target, respectively. In (1), the second equality is obtained by Taylor series expansion with respect to $\mathbf{r}_p$ up to the first order. $O(\mathbf{r}_p)$ includes high orders, which can be ignored under far-field assumption [31]. To facilitate the mathematical derivations in this article, the matrix variables are defined in the matrix form given in bold capital letters, the position variables are defined in the vector form given in bold small letters, and the distance variables are calculated in modulus of difference of vectors.

Commonly, SAR transmits linear frequency-modulated (LFM) pulses and receives the echoes to generate high-resolution SAR images. After range compression and azimuth dechirping, the echoed pulses can be expressed as [31]

$$S_0(k,t)=\sum_p \sigma_p \exp\left[-jk\left(\frac{R_0 r_p^r - v t r_p^x}{\sqrt{{R_0}^2+(vt)^2}}\right)\right]+cn \quad (2)$$

where σ_p stands for the backscattering coefficient of the pth scatters, cn denotes additive noise, under wavenumber denotations, the wavenumber variable can be given as $k = 4\pi f/\mathrm{c}$, where c is the speed of light, and f is the transmitting frequency variable. Then, SAR image formation processing (IFP) is required to remove the range migrations [32]. Considering the spotlight SAR, the polar formation algorithm (PFA) is applied, and the 2-D spectra after the PFA interpolation [31] in the wavenumber domain can be written as

$$S_0(k_r,k_x)=\sum_p \sigma_p \exp\left(-jk_x r_p^x - jk_r r_p^r\right)+cn \quad (3)$$

where $k_r = R_0 k/(R_0^2+v^2t^2)^{1/2}$ and $k_x = -vt'k_0/R_0$ are the range and azimuthal wavenumber variables, respectively. Both the variable substitutions can be implemented by linear interpolators [31]. In (3), t' is the resampled/interpolated slow-time variable [32]. It can be noted from (3) that the stationary scatters can be imaged at (r_p^r, r_p^x) by 2-D FT with respect to k_r and k_x, respectively. Applying 1-D FT in the range direction, the range compressed formation of (3) can be written as

$$S_0(\hat{r},t')=\sum_p \sigma_p \mathrm{sinc}\left(\hat{r}-r_p^r\right)\exp\left(j2\pi\frac{2vr_p^x}{\lambda R_0}t'\right)+cn \quad (4)$$

where $\mathrm{sinc}(\cdot)$ is the sinc function denoting the imaging response in the range direction. Setting f_0 to be the centroid frequency, the wavenumber centroid is $k_0 = 4\pi/\lambda$, where $\lambda = c/f_0$ is the wavelength.

SAR is originally designed for imagery of stationary scenarios. For moving targets, their responses will be shifted and defocused in the SAR image of the stationary scene. SAR-GMTIm aims to relocate and refocus the target image. As shown in Fig. 1, suppose the moving target is with a constant velocity $\mathbf{v}_l$. The slant range of moving targets can be shown as

$$\begin{aligned} R_l(t) &= |\mathbf{R}_0+\mathbf{r}_l+\mathbf{v}_l t-\mathbf{q}_0(t)| \\ &= |\mathbf{R}_0-\mathbf{q}_0(t)| + \frac{\mathbf{R}_0-\mathbf{q}_0(t)}{|\mathbf{R}_0-\mathbf{q}_0(t)|}(\mathbf{r}_l+\mathbf{v}_l t) \\ &\quad + \frac{(\mathbf{r}_l+\mathbf{v}_l t)^2}{2|\mathbf{R}_0-\mathbf{q}_0(t)|}(\mathbf{r}_l+\mathbf{v}_l t)^2 + O(\mathbf{r}_l+\mathbf{v}_l t) \end{aligned} \quad (5)$$

by Taylor series expansion with respect to $(\mathbf{r}_l+\mathbf{v}_l t)$ up to second order, where $\mathbf{r}_l$ gives the position offset vector of the moving target from the scene center O_c, and $O(\mathbf{r}_l+\mathbf{v}_l t)$ includes high orders that are negligible under far-field and narrow-beam assumptions [31]. To achieve ground moving target signatures in a high signal-to-noise ratio (SNR), the ground clutter is to be suppressed. Generally, multichannel SAR is employed, and displaced phase center antenna (DPCA) is applied [33] for clutter suppression. Similar to the derivation as in (4), after the PFA resampling, the clutter-suppressed moving target signal in the range-compressed data domain can be written as [32], [33]

$$\begin{aligned} S_{n-0}(\hat{r},t') = \sum_{l=1} & A_l \mathrm{sinc}\left(\hat{r}-r_l^r\right)\exp\left(j\frac{\phi_l}{2}\right) \\ & \times \exp\left(j2\pi f_d^l t' + j\pi \eta_d^l t'^2\right) + cn_{n-0} \end{aligned} \quad (6)$$

where cn_{n-0} denotes the residual unsuppressed clutter and thermal noise, and A_l and ϕ_l are the amplitude modulation and interferometric phase, respectively. It can be noted from (6) that due to the moving target, linear and quadratic phase modulations are introduced, where the Doppler centroid frequency and chirp rate f_d^l and η_d^l can be specifically derived as [32]

$$\begin{aligned} f_d^l &= \frac{2vr_l^x - 2\left(R_0+r_l^r\right)v_l^r}{\lambda R_0} \\ \eta_d^l &= \frac{-2\left[\left(v-v_l^x\right)^2+\left(v_l^r\right)^2\right]}{\lambda R_0} \end{aligned} \quad (7)$$

where v_l^x and v_l^r are the moving target velocity in azimuthal and slant-range directions, respectively, and $\mathbf{v}_l = v_l^x\bar{\mathbf{x}} + v_l^r\bar{\mathbf{r}}$.

In this article, we are to generate SG-SAR images for both stationary scenes and moving targets. By observing (4) and (6), both the range-compressed signal for stationary scene and moving target can be concisely written in matrices as

$$\mathbf{Y} = \mathbf{AX} + \mathbf{C_n} \quad (8)$$

where $\mathbf{Y} \in \mathbb{C}^{N\times M}$ denotes the range-compressed data matrix, and N and M represent the number of echoed pulses and range cells, respectively. For SAR imagery of stationary scenes, $\mathbf{Y}$ is $S_0(\hat{r},t')$ in (4), while $\mathbf{Y}$ is $S_{n-0}(\hat{r},t')$ in (6) for moving target imagery. In practice, motion deviations of the airborne platform and moving targets would be encountered. Auxiliary navigation units and/or adaptive data-driven autofocusing are necessary, which have been discussed in detail in our previous studies [33]. In (8), $\mathbf{C_n}$ gives the residual clutter and thermal noise in the range-compressed domain, and $\mathbf{A}$ denotes the representation dictionary that can be generated according to FT. When the common SAR imagery of a stationary scene is concerned, the Fourier dictionary can be given as

$$\begin{aligned} \mathbf{A}_0 &= \{\mathbf{a}[f_d(1)], \mathbf{a}[f_d(2)], \ldots, \mathbf{a}[f_d(N)]\} \\ \mathbf{a}[f_d(n)] &= \left[e^{-j2\pi f_d(n)t_1}, e^{-j2\pi f_d(n)t_2}, \ldots, e^{-j2\pi f_d(n)t_N}\right]^T \end{aligned} \quad (9)$$

and the intended $\mathbf{A} = \mathbf{A}_0$. While moving targets are to be focused, the Doppler chirp rates η_d^l should be considered. It can be estimated by time–frequency representation (TFR) methods, such as Lv's distribution (LVD) [34]. In such cases,

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$\mathbf{A}$ can be generated as

$$\begin{aligned}\mathbf{A} &= \mathbf{A}(\boldsymbol{\eta}_d) = [\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_l] \\ \mathbf{A}_l &= \mathbf{A}_0 \odot \mathbf{B}(\eta_d^l) \\ \mathbf{B}(\eta_d^l) &= \left[\mathbf{b}(\eta_d^l), \mathbf{b}(\eta_d^l), \ldots, \mathbf{b}(\eta_d^l)\right]_{1\times N} \\ \mathbf{b}(\eta_d^l) &= \left[e^{j\pi\eta_d^l t_1^2}, e^{j\pi\eta_d^l t_2^2}, \ldots, e^{j\pi\eta_d^l t_N^2}\right]^T \end{aligned} \tag{10}$$

where $\boldsymbol{\eta}_\mathbf{d} = [\eta_d^1, \eta_d^2, \ldots, \eta_d^l]$ includes Doppler chirp rates of the moving targets of interests, and $\odot$ is the Hadamard product operator. It can be observed from (10) that $\mathbf{A}$ is constructed by l elements. Each element $\mathbf{A}_l$ is generated under the Hadamard product. $\mathbf{A}_0$ is the common Fourier dictionary, and $\mathbf{B}$ is a quadratic dictionary to match filter the Doppler chirp rate η_d^l. Therefore, to achieve the focused image of multiple targets, $\mathbf{A}$ is inherently a quadratic combination of Fourier dictionary. According to the observation, the quadratic Doppler dictionary $\mathbf{A}(\boldsymbol{\eta}_d)$ in (10) will be degenerated into the common Fourier dictionary as in (9), when the velocity of the target is zero. In other words, the quadratic Doppler dictionary of the moving targets is also valid for stationary scenes. In such a case, the range-compressed signal for stationary scenes in (4) and moving targets in (6) can be summarized in (8). Finally, $\mathbf{X} \in \mathbb{C}^{N\times M}$ is the complex-valued SAR image or the intended features [33] to be enhanced.

III. Precursors

To facilitate the following discussions, reviews on conventional sparsity-driven SAR imagery and structure enhancement algorithms are carried out in this section. Also, the MM and generalized ADMM are introduced to establish fundamentals for the intended SG-SAR algorithm.

A. Conventional Sparsity-Driven SAR Imagery

Solving the inverse problem in (8) without $\mathbf{C_n}$ is a simple problem by matrix inversion. However, due to the residual clutter, additive noise, and under-sampled echoes, the inversion will be a typical ill-posed problem. To this end, encoding prior knowledge is an effective way. The ℓ_1 regularization is the most classic way to recover the sparse feature, where the optimization is usually modeled in an unconstrained LASSO form as [35]

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \left(\frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 + \zeta \|\mathrm{vec}(\mathbf{X})\|_1 \right) \tag{11}$$

where $\mathrm{vec}(\bullet)$ is the vectorized operator of a matrix, $\|\bullet\|_1$ is ℓ_1-norm denoting the sparse priori, $\zeta \geq 0$ is the penalty parameter for the constraint, and $\|\bullet\|_{\mathrm{F}}$ is Frobenius-norm named by data fidelity measuring the recovering accuracy. Although the sparsity-driven SAR imagery shows superior performance under low SNR and undersampling circumstances, it is incapable of preserving the structure feature of the target of interest. Since the sparse recovery only highlights dominant features, weak scattering features are easily lost. It is difficult to extract the elaborate structural features. The imaging accuracy will be limited.

B. Conventional Structure Enhancement Algorithms

To realize the enhancement of clustering features in the SAR imagery, structural priors are designed and developed under regularization [36] and Bayesian frameworks [35], respectively.

1) Overlapping Group LASSO: An overlapping group LASSO (OGL) is proposed in [17], which is an extended version of LASSO by replacing the ℓ_1-norm with a flexible ℓ_1/ℓ_2-mixed norm. The formulation for the OGL model can be shown as [17]

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \left(\frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 + \gamma \sum_{i=1}^{G} \omega_i \|\mathbf{X}_i\|_{\mathrm{F}} \right) \tag{12}$$

where γ is the penalty parameter for the ℓ_1/ℓ_2 norm, ω_i is the weight for the ith group, $\mathbf{X}_i$ represents the ith group of the target priori, and $\mathbf{X} = \{\mathbf{X}_1, \ldots, \mathbf{X}_G\}$. However, the OGL still has some limitations, where the ℓ_1/ℓ_2-mixed norm treats each group as the smallest sparse unit to achieve outer group sparsity. Therefore, it is impossible to achieve sparsity within the group.

2) Sparse Group LASSO: To break the limitations of the OGL model, an SGL model is proposed in [19], which enables the sparsity of both inside and outside the groups, simultaneously. The problem for the SGL model can be formulated as [19]

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \left(\frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 + \gamma_1 \sum_{i=1}^{G} \|\mathbf{X}_i\|_{\mathrm{F}} + \gamma_2 \|\mathrm{vec}(\mathbf{X})\|_1 \right) \tag{13}$$

where an additional ℓ_1-norm is introduced, and γ_1 and γ_2 are the penalty parameters for ℓ_1/ℓ_2-mixed norm and ℓ_1-norm, respectively. Although the SGL improves the performance of structure feature enhancement, it leads to a complicated tuning of multiple parameters [37]. Besides, considering the rich content of the ground scenario, the practical structure of the target to be imaged would be nonhomogeneous. Therefore, both OGL and SGL would not be flexible enough to capture the complicated features to be enhanced.

3) Block-Sparsity Bayesian Learning: The structure enhancement can also be achieved by statistical Bayesian learning. The mathematical formation can be shown as [32]

$$\hat{\mathbf{X}} = \langle p(\mathbf{X}|\mathbf{Y}) \rangle, \quad \text{s.t. } p(\mathbf{X}|\mathbf{Y}) = \frac{p(\mathbf{X})p(\mathbf{Y}|\mathbf{X})}{p(\mathbf{Y})} \tag{14}$$

where $\langle \bullet \rangle$ denotes the expectation operator, $\hat{\mathbf{X}}$ is the estimation, $p(\mathbf{X}|\mathbf{Y})$ represents the posterior probability density distribution of $\mathbf{X}$, $p(\mathbf{X})$ is the prior distribution, and $p(\mathbf{Y})$ is the integral of the numerator in (14). To achieve a closed-form solution for the posterior, a hierarchical Bayesian model is applied [32]. However, the complexity of the hierarchical formation increases dramatically followed by the block sparsity before being encoded, and the solvability of the problem will be questioned. Besides, the computational load will become large, which limits the practicability of Bayesian learning.

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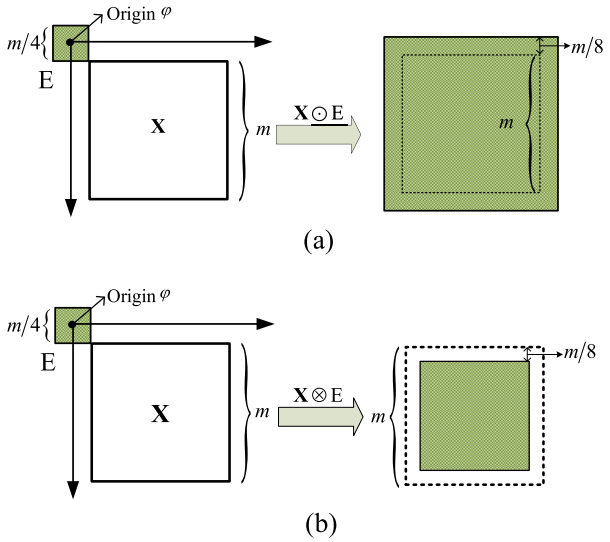


Fig. 2. Dilation and erosion diagram of fixed SE after traversal. (a) Dilation. (b) Erosion.

C. Mathematical Morphology

The classical MM [25] is a powerful tool in image processing, which is capable of introducing topological and geometric continuity to quantitatively describe the structure, such as size, shape, convexity, connectivity, geodesic distance, and so on [38]. Generally, MM consists of a set of operators such as dilation, erosion, opening, and closing. The dilation and erosion operators for conventional morphology can be written as

$$\begin{cases} \mathbf{X} \odot E = \left\{ \varphi \middle| \left[(E)_\varphi \cap \mathbf{X} \right] \neq \emptyset \right\} = \underset{e \in E}{\cup} (\mathbf{X})_e \\ \mathbf{X} \otimes E = \left\{ \varphi \middle| (E)_\varphi \subseteq \mathbf{X} \right\} = \underset{e \in E}{\cap} (\mathbf{X})_{-e} \end{cases} \tag{15}$$

where $\mathbf{X}$ denotes the interested target image/feature, $\odot$ and $\otimes$ denote the dilation and erosion operators, respectively, E is the SE with a fixed shape, φ is the origin of the SE, and $e \in E$ represents the union of all feasible values after the traversal of fixed SE. When E does not intersect with $\mathbf{X}$ in an empty set, the set of all the positions of origin φ constructs the dilation image. While E is completely contained in $\mathbf{X}$, the set of the origin positions of E construct the erosion image. It can be noted from (15) that the dilation operator is to merge the origin φ of SE into $\mathbf{X}$ as long as SE and $\mathbf{X}$ are connected, so as to form the dilation after the traversal. On the contrary, when SE is completely contained in $\mathbf{X}$, the origin φ is retained, and an erosion image smaller than the original $\mathbf{X}$ is formed. Fig. 2(a) and (b) shows the dilation and erosion diagrams after traversing $\mathbf{X}$ with a fixed length of $m/4$ in square SE, respectively. By defining the subgradients of operators, morphological regularization (MR) is constructed. Complicated structures can be captured under the MR. Therefore, the MM has great potential in terms of structure enhancement in high-resolution SAR imagery.

However, there are some drawbacks to conventional MR. First, it is common to adopt fixed SE with limited size. The smaller SE, the poorer the noise suppression capability. The larger SE, the poorer the integrity of target structures. Therefore, a fixed SE will lead to inflexibility in practice. Second, due to the nonconvexity of the conventional MR, its convergence performance cannot be guaranteed. Finally, only a single task of MR is poor in super-resolution and denoising, and it is necessary to incorporate additional priors to enhance multiple features in a multitask learning manner.

D. Generalized Alternating Direction Method of Multipliers (ADMM)

There are plenty of convex optimization methods, such as greedy iteration (GI), basis pursuit (BP), and so on [33]. However, they are poor in coordinating multiple features to be enhanced. The ADMM [39] develops as a simple but powerful convex optimization method. It is capable of optimizing multiple regularizers and solving the joint optimization in a multitask manner. To accommodate general applications, a generalized ADMM formation is established to enhance multiple features in a constrained formed as

$$\arg\min_{\mathbf{X}} \sum_h \gamma_h g_h(\mathbf{X}), \quad \text{s.t. } \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 \leq \tau \tag{16}$$

where τ is an empirical threshold for the accuracy of feature recovery, $g_h(\mathbf{X})$ denotes the objective function of the hth feature to be enhanced, and γ_h is regularization parameter. According to the ADMM procedure, the joint problem in (16) can be decomposed into subproblems, and the variable splitting scheme can be carried out as

$$\min f(\mathbf{X}) + g(\mathbf{Z}), \quad \text{s.t. } \mathbf{B}\mathbf{X} + \mathbf{D}\mathbf{Z} = \mathbf{W} \tag{17}$$

where $f(\mathbf{X}) = (1/2)\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2$ is the fidelity term to measure the accuracy, $\mathbf{Z}$ is the splitted variable given as $g(\mathbf{Z}) = \sum_{h=1} \gamma_h g_h(\mathbf{Z}_h)$ with multiple regularizers $g_h(\mathbf{Z}_h)$ and splitted variables $\mathbf{Z}_h$, and there is $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_{\mathrm{H}}]^T$. Next, the original problem is decomposed into several subproblems. In such cases, the augmented Lagrangian form of (17) can be given as

$$L_\rho(\mathbf{X}, \mathbf{Z}, \mathbf{U}) = f(\mathbf{X}) + \mathrm{g}(\mathbf{Z}) + \mathbf{U}^{\mathrm{H}}(\mathbf{B}\mathbf{X} + \mathbf{D}\mathbf{Z} - \mathbf{W}) + \frac{\rho}{2}\|\mathbf{B}\mathbf{X} + \mathbf{D}\mathbf{Z} - \mathbf{W}\|_{\mathrm{F}}^2 \tag{18}$$

where $\mathbf{U}$ is the dual variable, ρ is the penalty parameter, and the superscript $\{\cdot\}^{\mathrm{H}}$ is the conjugate transpose operator. Then, according to the coordinate descent mechanism, the solution in the scaled form can be derived as [33]

$$\begin{cases} \textbf{for } h = 1 : H \\ \quad \mathbf{Z}_h^{k+1} = \arg\min_{\mathbf{Z}_h} \left(g_h(\mathbf{Z}_h) + \frac{\rho}{2} \left\| \mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}_h - \mathbf{S}^k \right\|_{\mathrm{F}}^2 \right) \\ \quad \mathbf{S}_h^{k+1} = \mathbf{S}_h^k - \mathbf{B}\mathbf{X}^k - \mathbf{D}\mathbf{Z}_h^{k+1} \\ \textbf{end} \\ \mathbf{Z}^{k+1} = \left[\mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1} \ldots \mathbf{Z}_H^{k+1} \right] \\ \mathbf{S}^{k+1} = \left[\mathbf{S}_1^{k+1}, \mathbf{S}_2^{k+1} \ldots \mathbf{S}_H^{k+1} \right] \\ \mathbf{X}^{k+1} = \arg\min_{\mathbf{X}} \left(f(\mathbf{X}) + \frac{\rho}{2} \left\| \mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1} \right\|_{\mathrm{F}}^2 \right) \end{cases} \tag{19}$$

where $\mathbf{S}^{k+1} = (1/\rho)\mathbf{U}^{k+1}$ is the scaled dual variable and k denotes the iteration index. From (19), in the alternative update

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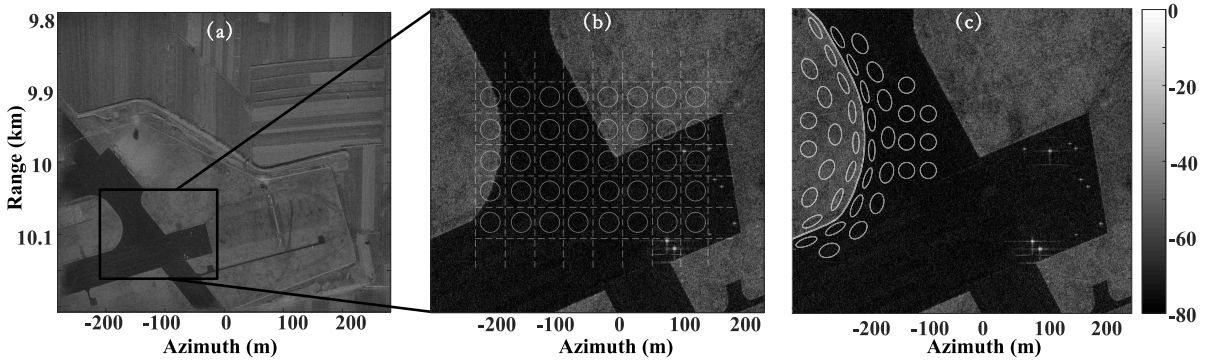


Fig. 3. Schematic of spatially variant SE. (a) SAR image. (b) SE of conventional morphology. (c) SE of spatially variant morphology.

of ADMM, $\mathbf{Z}_h^{k+1}$ and its corresponding dual variables $\mathbf{S}_h^{k+1}$ are first updated as distributed learning. Then, $\mathbf{X}^{k+1}$ is estimated by combining the results from the distributions, and it is jointly updated in a ridge regression manner. The variable $\mathbf{Z}$ is split into H feature variables to achieve multifeature enhancement. Therefore, the generalized ADMM is well-suited for multitask learning [33]. In this article, we are to develop the intended SG-SAR algorithm in the generalized ADMM framework, so that both the structural and sparse features can be enhanced in a fully cooperative manner.

IV. SG-SAR Imaging Algorithm

In this article, we are to design a novel morphology norm and implement the regularization under the generalized ADMM framework. Due to the cooperative framework, both the sparse and structural priors are taken into account, and both the features can be highlighted synergistically. Superior convergence and balanced performance between accuracy and efficiency can be guaranteed. Specifically, two splitted/feature variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are introduced, and the intended imagery result $\mathbf{X}$ will be coordinated under the splitted variables and the corresponding dual variables $\mathbf{S}_1$ and $\mathbf{S}_2$. According to (17)–(19) and considering characteristics of the morphology and sparse regularizers, we set the equalities $\mathbf{B} = [\mathbf{I}, \mathbf{I}]^T$, $\mathbf{D} = -\mathbf{I}$ and $\mathbf{W} = 0$, and we have $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{X}$.

A. Structure Enhancement With Spatially Variant Morphology Regularizer

In this section, a spatially variant morphology regularizer is incorporated within the generalized ADMM, and the structural feature enhancement can be implemented. First, by exploiting the structure tensor, the spatially variant SE is designed. Next, according to the designated SE, the morphology norm can be determined, and the intended proximity operator of the morphological regularizer will be derived in an analytical form. The clustering feature of the target of interest can be accordingly highlighted. Specifically, the splitted variable $\mathbf{Z}_1$ is introduced for the structural feature to be enhanced.

1) Spatially Variant Morphology Regularizer: According to the structure tensor theory, the specific definition of the spatially variant SE can be given as

$$E_{\mathrm{sv}}[\alpha(x), \beta(x), \Omega(x)] \tag{20}$$

where $x \in \mathbf{X}$ is an arbitrary resolution cell in the image data $\mathbf{X}$. In (20), the intended SE is to be implemented with an auto-fitting manner, which is desired to continently vary with the structure of the scatterer. More specifically, an elliptic SE is to be designed, and α, β, and Ω denote the semimajor axis, semiminor axis, and direction of the SE, respectively. Their specific expressions can be shown as

$$\begin{cases} \alpha(x) = \dfrac{(\varepsilon_1(x) + K) \times D}{\varepsilon_1(x) + \varepsilon_2(x) + 2K} \\ \beta(x) = \dfrac{(\varepsilon_2(x) + K) \times D}{\varepsilon_1(x) + \varepsilon_2(x) + 2K} \\ \Omega(x) = \begin{cases} \arctan\left(\dfrac{\eta_2^r(x)}{\eta_2^a(x)}\right), & \eta_2^a(x) \neq 0 \\ \pi 2, & \eta_2^a(x) = 0 \end{cases} \end{cases} \tag{21}$$

where K is a small constant and D is the maximum length of the semimajor axis defined by $0 \leq \beta(x) \leq \alpha(x) \leq D$. $\varepsilon_1(x)$ and $\varepsilon_2(x)$ represent the eigenvalues of each resolution cell x in the azimuth and range, respectively. $\eta_2^a(x)$ and $\eta_2^r(x)$ are the components of eigenvector $\eta_2(x)$ in the azimuth and range directions, respectively. $\Omega(x)$ is the angle between $\alpha(x)$ and the azimuth. It can be noted from (21) that the elliptic SE will become straight lines as the semiminor axis $\beta(x)$ approaches zero in the scenario with strong anisotropic scatters, where $\Omega(x) = \pi/2$ and $\varepsilon_1(x) \gg \varepsilon_2(x)$. On the contrary, the elliptic SE will become a circle in the isotropic region, where $\varepsilon_1(x) \approx \varepsilon_2(x)$, that is, $\alpha(x) \approx \beta(x)$.

Fig. 3 explicitly illustrates the intended spatially variant SE compared with conventional morphology. In conventions, all the scatters within the scene of interest are equivalently treated, and the SE is in isotropic circles that are uniformly distributed, as shown in Fig. 3(b) enlarged from Fig. 3(a). While in the intended spatially variant SE, anisotropic ellipses are incorporated as shown in Fig. 3(c), which are with proper directions that accommodate the specific structures. To obtain the intended parameters of the elliptic SE, a structure tensor is applied, and the intended eigenvalues and the corresponding eigenvectors can be achieved from the definition of the structure tensor of the intended target $\mathbf{X}$ as

$$J_\sigma(\mathbf{X}) = G_\sigma * \left[\nabla \mathbf{X}(x) \nabla^{\mathrm{T}} \mathbf{X}(x)\right] \tag{22}$$

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where G_σ is a Gaussian convolution kernel of standard deviation σ, $*$ denotes convolution operator, ∇ is the gradient operator, which can be defined as $\nabla = ((\partial/\partial x_a), (\partial/\partial x_r))^T$, and $(x_a, x_r)^T$ is the coordinates of x in the slant-range plane. The structure tensor $J_\sigma(\mathbf{X})$ in (22) is a symmetric and positive semidefinite matrix, which can be further applied for the eigenvalue decomposition. Therefore, the eigenvalues $\varepsilon_1(x)$ and $\varepsilon_2(x)$ in (21) can be obtained, and the spatially variant SE in ellipse shape can be generated as in (20) for the intended morphology regularizer.

From the intended SE definition in (20), the spatially variant morphological operators can be constructed as

$$\begin{cases} \omega_{E_{\text{sv}}}(\mathbf{X}) = \underset{x' \in E_{\text{sv}}(x)}{\vee} \mathbf{X}(x') \\ \varpi_{E_{\text{sv}}}(\mathbf{X}) = \underset{x' \in E_{\text{sv}}(x)}{\wedge} \mathbf{X}(x') \\ \delta_{E_{\text{sv}}}(\mathbf{X}) = \varpi_{E_{\text{sv}}}\left(\omega_{E_{\text{sv}}}(\mathbf{X})\right) \\ \kappa_{E_{\text{sv}}}(\mathbf{X}) = \omega_{E_{\text{sv}}}\left(\varpi_{E_{\text{sv}}}(\mathbf{X})\right) \end{cases} \tag{23}$$

where ω and ϖ denote morphological dilation and erosion operators, respectively, and $\vee$ and $\wedge$ are the maximum and minimum operators, respectively. x' is the variable within the domain of the SE whose origin is x. During the dilation operation, the internal maximum value of the SE is solved to assign the value to the origin, while the erosion is just the opposite to obtain the minimum value. In (23), δ and κ denote spatially variant morphological closing and opening operators, respectively.

Finally, according to the elliptic SE as designed in (20), the spatially variant morphology regularizer can be derived by substituting (23) into the $\mathbf{Z}_1$ regularization of (19) as

$$g_1(\mathbf{Z}_1) = \mathbf{Z}_1 \sum_{D=1} m\left[\delta_{E_{\text{sv}}}(\mathbf{Z}_1) - \kappa_{E_{\text{sv}}}(\mathbf{Z}_1)\right] \tag{24}$$

where m is the weighting coefficient, D satisfies $D \propto \alpha/\beta \in E_{\text{sv}}$, and the variation in D is accompanied by the change in SE E_{sv}. In (24), $m[\delta_{E_{\text{sv}}}(\mathbf{Z}_1) - \kappa_{E_{\text{sv}}}(\mathbf{Z}_1)]$ performs as a mask that is capable of accommodating to the complex-valued SAR data, which improves the applicability of the proposed SG-SAR algorithm. Moreover, the closing operator in (24) connects the detailed features, the opening operator eliminates the speckles, and the subtraction finally enhances the useful structures of the targets of interest.

2) Proximity Operator of Spatially Variant Morphology Regularizer: Under the designated spatially variant morphology regularizer as shown in (24), the solution of the structural feature will be obtained from (19) by solving a typical proximal minimization problem as

$$\mathbf{Z}_1^{k+1} = \arg\min_{\mathbf{Z}_1}\left(\lambda_1 g_1(\mathbf{Z}_1) + \frac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_1 - \mathbf{S}_1^k\right\|_{\text{F}}^2\right) \tag{25}$$

where λ_1 is the regularization coefficient. Because of the nondifferentiable of the regularizer in (24), (25) is difficult to achieve a closed-form solution. To this end, the proximal algorithm is applied, and the proximal mapping of the intended morphological regularizer can be derived as

$$\mathbf{Z}_1^{k+1} = \text{prox}_{\lambda_1/\rho}\left(\mathbf{X}^k - \mathbf{S}_1^k\right) = \frac{\partial \mathbf{W}^{k+1}}{\partial \mathbf{W}}$$

$$\begin{aligned} &= \frac{\partial}{\partial \mathbf{W}} \sum_{D=1} m\left[\delta_{E_{\text{sv}}}(\mathbf{W}) - \kappa_{E_{\text{sv}}}(\mathbf{W})\right] \\ &= \sum_{D=1} m\left\{\frac{\partial}{\partial \mathbf{W}}\left[\varpi_{E_{\text{sv}}}\left(\omega_{E_{\text{sv}}}(\mathbf{W})\right)\right]\right. \\ &\qquad \left. - \frac{\partial}{\partial \mathbf{W}}\left[\omega_{E_{\text{sv}}}\left(\varpi_{E_{\text{sv}}}(\mathbf{W})\right)\right]\right\} \end{aligned} \tag{26}$$

where $\text{prox}_{\lambda_1/\rho}(\bullet)$ denotes the proximity operator of MR, and $\mathbf{X}^k - \mathbf{S}_1^k = \mathbf{W}^{k+1}$. In (26), to accommodate the nondifferentiable morphological regularizer, subgradient optimization is carried out. By substituting the intended morphological regularizer in (24) into the subgradient equality in (26), the solution takes the difference between the closing operator and the opening operator to find the derivative. Furthermore, the specific definitions of $\omega_{E_{\text{sv}}}(\mathbf{W})$ and $\varpi_{E_{\text{sv}}}(\mathbf{W})$ in (26) are

$$\begin{cases} \omega_{E_{\text{sv}}}(\mathbf{W}) \\ = \begin{bmatrix} \max(x_i)_{i \in E_{\text{sv}(1\times 1)}} & \cdots & \max(x_i)_{i \in E_{\text{sv}(1\times M)}} \\ \vdots & \ddots & \vdots \\ \max(x_i)_{i \in E_{\text{sv}(N\times 1)}} & \cdots & \max(x_i)_{i \in E_{\text{sv}(N\times M)}} \end{bmatrix} \\ \varpi_{E_{\text{sv}}}(\mathbf{W}) \\ = \begin{bmatrix} \min(x_i)_{i \in E_{\text{sv}(1\times 1)}} & \cdots & \min(x_i)_{i \in E_{\text{sv}(1\times M)}} \\ \vdots & \ddots & \vdots \\ \min(x_i)_{i \in E_{\text{sv}(N\times 1)}} & \cdots & \min(x_i)_{i \in E_{\text{sv}(N\times M)}} \end{bmatrix} \end{cases} \tag{27}$$

where the dilation operator $\omega_{E_{\text{sv}}}$ is equivalent to assigning the maximum value of the SE to its origin. On the contrary, the erosion operator $\varpi_{E_{\text{sv}}}$ is given by the minimum value. According to (27), the definition of $(\partial/\partial \mathbf{W})[\varpi_{E_{\text{sv}}}(\omega_{E_{\text{sv}}}(\mathbf{W}))]$ and $(\partial/\partial \mathbf{W})[\omega_{E_{\text{sv}}}(\varpi_{E_{\text{sv}}}(\mathbf{W}))]$ in (26) can be specifically derived as

$$\begin{cases} \dfrac{\partial\left[\varpi_{E_{\text{sv}}}\left(\omega_{E_{\text{sv}}}(\mathbf{W})\right)\right]}{\partial \mathbf{W}} = \dfrac{\partial \omega_{E_{\text{sv}}}(\mathbf{W})}{\partial \mathbf{W}} \dfrac{\partial\left[\varpi_{E_{\text{sv}}}\left(\omega_{E_{\text{sv}}}(\mathbf{W})\right)\right]}{\partial \omega_{E_{\text{sv}}}(\mathbf{W})} \\ \qquad = \begin{cases} 1, & r_i = \max(x_i)_{i \in E_{\text{sv}(N\times M)}} \\ & \text{and } \omega_i = \min(\omega_i)_{i \in E_{\text{sv}(N\times M)}} \\ 0, & \text{others} \end{cases} \\ \dfrac{\partial\left[\omega_{E_{\text{sv}}}\left(\varpi_{E_{\text{sv}}}(\mathbf{W})\right)\right]}{\partial \mathbf{W}} = \dfrac{\partial \varpi_{E_{\text{sv}}}(\mathbf{W})}{\partial \mathbf{W}} \dfrac{\partial\left[\omega_{E_{\text{sv}}}\left(\varpi_{E_{\text{sv}}}(\mathbf{W})\right)\right]}{\partial \varpi_{E_{\text{sv}}}(\mathbf{W})} \\ \qquad = \begin{cases} 1, & r_i = \min(x_i)_{i \in E_{\text{sv}(N\times M)}} \\ & \text{and } \varpi_i = \min(\varpi_i)_{i \in E_{\text{sv}(N\times M)}} \\ 0, & \text{others} \end{cases} \end{cases} \tag{28}$$

where $\omega_i \in \omega_{E_{\text{sv}}}(\mathbf{W})$ and $\varpi_i \in \varpi_{E_{\text{sv}}}(\mathbf{W})$. Substituting (28) into (26), the solution for the structure feature can be obtained in the specific form. The final solution for the spatially variant morphology regularizer can be found in (26).

As mentioned above, the proposed SG-SAR algorithm utilizes the spatially variant morphology regularizer to achieve a structural enhanced image, where an adaptive elliptic SE is designed to accommodate various structures. Fig. 4 illustrates the differences among various structure metrics, such as grouped structure, MM, and so on. For instance, there is a target with a complicated structure, where the spatial distance measures the structure in orthogonal coordinates, and the Euclidean distance measures the straight structure as the conventional grouping scheme [40]. The spatially variant MR

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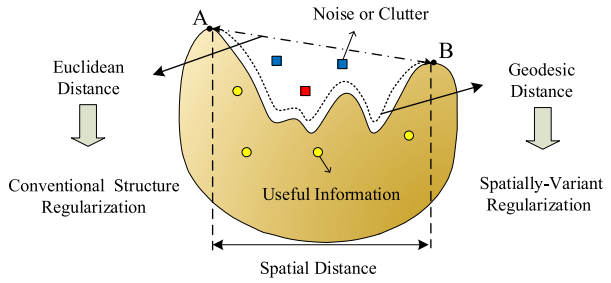


Fig. 4. Schematic of the proposed morphological regularizer.

optimizes in the geodesic distance as the dashed line shown in the outline of Fig. 4. The geodesic distance is a kind of manifold distance that is capable of measuring the intrinsic structure of scatters' clustering in the SAR imagery. Therefore, the intended regularizer can perfectly auto-fit the outline of the target, which can retrieve the useful structure features and avoid the additive noise and clutter beyond the target.

To evaluate the accuracy of the feature learning for the variable $\mathbf{Z}_1$, a scaled dual-variable $\mathbf{S}_1$ is introduced and synchronously updated as

$$\mathbf{S}_1^{k+1} = \mathbf{X}^k - \mathbf{Z}_1^{k+1} + \mathbf{S}_1^k \tag{29}$$

where the dual-variable can be viewed as a residual between the primal variable $\mathbf{X}$ and the splitted variable $\mathbf{Z}_1$.

B. Sparsity Enhancement With ℓ_1-Norm Regularizer

Based on the structure-guaranteed MR under spatially variant SE, a further sparse feature enhancement is to be incorporated. To this end, the sparse regularization is to be encoded in a collaborative manner. Specifically, the MR and the sparse regularization are developed within the ADMM framework in a multitask learning scheme. In such cases, an additional splitted variable $\mathbf{Z}_2$ is introduced to represent the sparse feature to be enhanced, and the intended multitask learning will be operated in a "distributed-cooperative" manner.

The ℓ_1 norm is encoded for the sparse feature of the splitted variable $\mathbf{Z}_2$. According to the coordinate decent scheme in (19), the optimization for the sparsity enhancement can be formulated in the LASSO model as

$$\mathbf{Z}_2^{k+1} = \arg\min_{\mathbf{Z}_2}\left(\lambda_2\|\mathbf{Z}_2\|_1 + \frac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_2 - \mathbf{S}_2^k\right\|_{\mathrm{F}}^2\right). \tag{30}$$

In order to obtain a closed-form solution to the minimization problem (30), the proximal algorithm [33] is applied for the complex-valued SAR data, where a complex soft-thresholding operator can be derived as

$$\begin{aligned}\mathbf{Z}_2^{k+1} &= \mathrm{prox}_{\lambda_2/\rho}\left(\mathbf{X}^k + \mathbf{S}_2^k\right)\\ &= \left\{\left(\mathbf{X}^k + \mathbf{S}_2^k\right) - \frac{\lambda_2/\rho\left(\mathbf{X}^k + \mathbf{S}_2^k\right)}{\left|\mathbf{X}^k + \mathbf{S}_2^k\right|}\right\}_+\\ &= \frac{\max\left[\left(\left|\mathbf{X}^k + \mathbf{S}_2^k\right| - \lambda_2/\rho\right), 0\right]}{\max\left[\left(\left|\mathbf{X}^k + \mathbf{S}_2^k\right| - \lambda_2/\rho\right), 0\right] + \lambda_2/\rho}\left(\mathbf{X}^k + \mathbf{S}_2^k\right)\end{aligned} \tag{31}$$

where $\mathrm{prox}_{\lambda_2/\rho}(\bullet)$ denotes the complex soft-thresholding operator and $|\bullet|$ is the module operator. $\mathbf{S}_2$ represents the sparse dual-variable that can be updated as

$$\mathbf{S}_2^{k+1} = \mathbf{X}^k - \mathbf{Z}_2^{k+1} + \mathbf{S}_2^k \tag{32}$$

where the dual-variable $\mathbf{S}_2^{k+1} - \mathbf{S}_2^k$ can also be viewed as a residual between the primal and split variables.

C. Cooperative Learning for Joint Feature Enhancement

To achieve the intended feature-enhanced image result, cooperative learning is performed, where both the feature solutions of structure and sparsity are coordinated. The proposed algorithm is carried out in a "distributed-cooperative" manner, where the morphological and sparse regularizations perform as the distributed learning, and the cooperative learning is implemented for the primal variable $\mathbf{X}$ and updated as

$$\begin{aligned}\mathbf{X}^{k+1} = \arg\min_{\mathbf{X}}\Bigg(&\frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2\\ &+ \frac{\rho}{2}\left\|\mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right\|_{\mathrm{F}}^2\Bigg)\end{aligned} \tag{33}$$

where $\mathbf{Z}$ denotes the combined splitted variable as $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2]$, and $\mathbf{S}$ is the combined dual-variable as $\mathbf{S} = [\mathbf{S}_1, \mathbf{S}_2]$. Next, the objective function in (33) can be further written as

$$\begin{aligned}\Phi(\mathbf{X}) &= \frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 + \frac{\rho}{2}\left\|\mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right\|_{\mathrm{F}}^2\\ &= \frac{1}{2}\mathrm{Tr}\left[(\mathbf{Y} - \mathbf{A}\mathbf{X})^{\mathrm{H}}(\mathbf{Y} - \mathbf{A}\mathbf{X})\right] + \frac{\rho}{2}\\ &\quad\times \mathrm{Tr}\Big[\left(\mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right)^{\mathrm{H}}\\ &\quad\times\left(\mathbf{B}\mathbf{X}^k + \mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right)\Big]\end{aligned} \tag{34}$$

where $\mathrm{Tr}[\bullet]$ is the trace operator of a matrix. From (34), it is obviously a convex function, and the minima can be found when the derivative of the objective function achieves as $((d\Phi(\mathbf{X}))/d\mathbf{X}) = 0$, where

$$\begin{aligned}\frac{d\Phi(\mathbf{X})}{d\mathbf{X}} &= \mathbf{A}^{\mathrm{H}}\mathbf{A}\mathbf{X} - \mathbf{A}^{\mathrm{H}}\mathbf{Y} + \rho\mathbf{B}^{\mathrm{H}}\mathbf{B}\mathbf{X}^k\\ &\quad + \rho\mathbf{B}^{\mathrm{H}}\left(\mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right).\end{aligned} \tag{35}$$

The optimal value of $\mathbf{X}$ can be therefore solved as

$$\begin{aligned}\mathbf{X}^{k+1} &= \left(\mathbf{A}^{\mathrm{H}}\mathbf{A} + \rho\mathbf{I}\right)^{-1}\left[\mathbf{A}^{\mathrm{H}}\mathbf{Y} - \rho\mathbf{B}^{\mathrm{H}}\left(\mathbf{D}\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right)\right]\\ &= \left(\mathbf{A}^{\mathrm{H}}\mathbf{A} + \rho\mathbf{I}\right)^{-1}\left[\mathbf{A}^{\mathrm{H}}\mathbf{Y} + \rho_1\left(\mathbf{Z}_1^{k+1} + \mathbf{S}_1^{k+1}\right)\right.\\ &\quad\left. + \rho_2\left(\mathbf{Z}_2^{k+1} + \mathbf{S}_2^{k+1}\right)\right]\end{aligned} \tag{36}$$

where ρ_1 and ρ_2 are introduced for the penalty parameters of $\mathbf{Z}_1$ and $\mathbf{Z}_2$, respectively. From (36), the cooperative learning coordinates both the feature/splitted variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$, and the residual/dual-variables $\mathbf{S}_1$ and $\mathbf{S}_2$, synergistically and analytically.

V. Processing Flowchart

To summarize the processing procedure of the proposed SG-SAR algorithm, Fig. 5 is given to demonstrate the flowchart. Generally, the proposed algorithm aims at the raw SAR data, and preprocessing steps, such as IFP [41] and parameter

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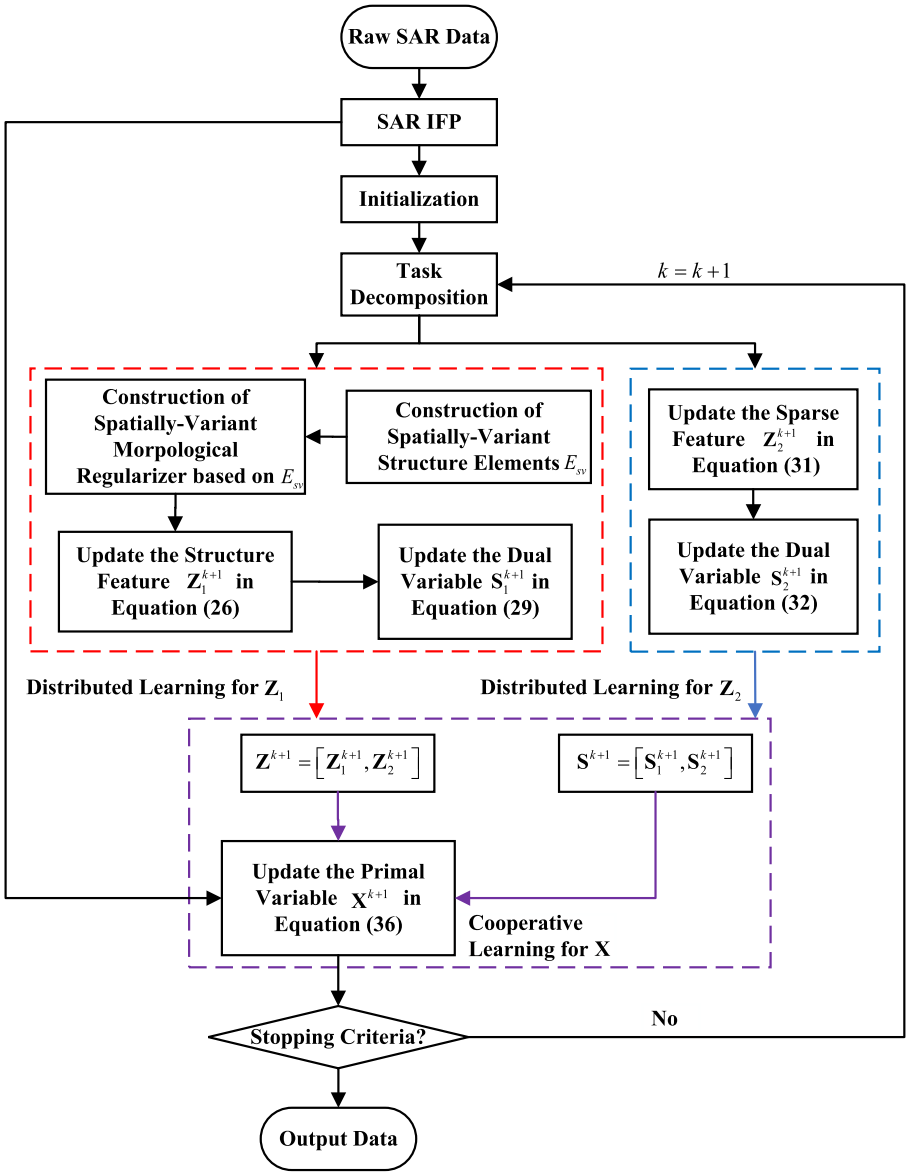


Fig. 5. Processing flowchart of the proposed SG-SAR algorithm.

initialization should be carried out before the multitask learning framework. Under a certain task decomposition scheme, the multitask learning works with three main steps, including two distributed learning for $\mathbf{Z}_1$, $\mathbf{S}_1$ and $\mathbf{Z}_2$, $\mathbf{S}_2$ respectively, and one cooperative learning for $\mathbf{X}$.

1) *Distributed Learning for $\mathbf{Z}_1$:* The first distributed learning introduces spatially variant SE and the corresponding operator to build the regularizer for $\mathbf{Z}_1$. Then, updating $\mathbf{Z}_1$ by solving its proximal mapping in (26) to achieve the structure feature enhancement. The related dual-variable $\mathbf{S}_1$ is updated as in (29).
2) *Distributed Learning for $\mathbf{Z}_2$:* The second distributed learning achieves sparse feature enhancement. The sparse enhancement is fulfilled by solving the ℓ_1-norm regularization as in (31), where a closed-form solution in a soft-thresholding operator is obtained. Also, the related dual-variable $\mathbf{S}_2$ is updated as in (32).
3) *Cooperative learning for X:* Under the obtained solutions for the structural feature $\mathbf{Z}_1$ and sparse feature $\mathbf{Z}_2$, and the solutions for the residuals $\mathbf{S}_1$ and $\mathbf{S}_2$, the primal imaging result $\mathbf{X}$ is coordinated by combining the splitted variables and dual-variables first as shown in (34), and then the cooperative learning is implemented as the closed-form solution as shown in (36). The stopping criteria for the iteration are defined as a residual variable r_k, and the threshold of the stopping criteria is fixed. The residual at iteration k is $r_k = \mathbf{B}\mathbf{X}^{k+1} - \mathbf{D}\mathbf{Z}^k$. The maximum residual value is set to 10^{-5}. If the residual value is less than or equal to 10^{-5}, the final output of the image will be $\mathbf{X}$ with enhanced structural and sparse features simultaneously. If else, the iteration continues. From the processing procedure in Fig. 5, the proposed SG-SAR algorithm is carried out as a multitask learning scheme in the "distributed-cooperative" learning manner.

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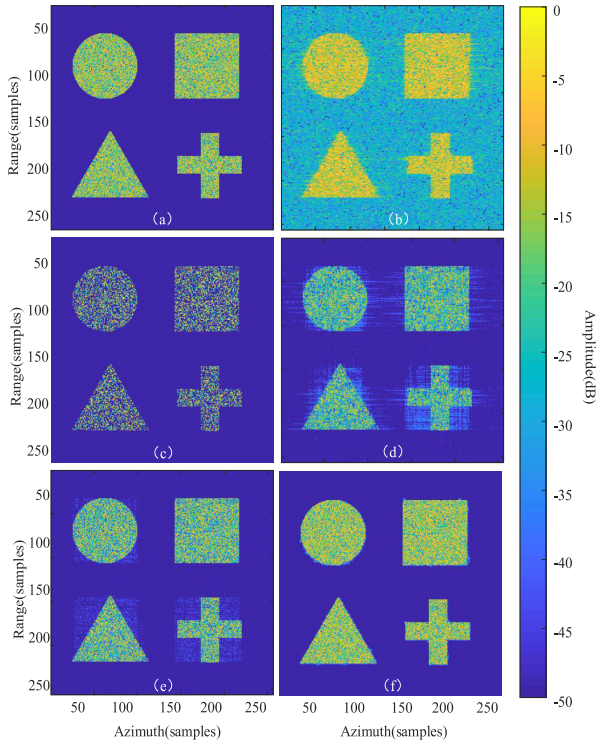


Fig. 6. Simulation results. (a) Reference image. (b) Direct FT image. (c) LASSO-ADMM image. (d) OGL-ADMM image. (e) SGL-ADMM image. (f) Proposed SG-SAR image.

VI. Experiments

In this section, experimental results will be reported to illustrate the performance of the proposed SG-SAR algorithm. Comparisons with conventional algorithms will be performed to demonstrate the superiority of the proposed algorithm. Both simulated and raw data are employed for the verification in terms of qualitative and quantitative aspects.

A. Simulated Data Experiments

In this section, polymorphic simulations based on complex-valued data are carried out to verify the superiority of the proposed algorithm over the conventional algorithms. The results are given in Fig. 6. Ground truth for the simulation is generated by randomly simulating scatters within polymorphic targets as shown in Fig. 6(a). Fig. 6(b) is the result generated by FT under the conditions of the undersampling ratio (USR) of 70% and Gaussian white noise of 5 dB SNR. Due to the contamination, the result is with high sidelobes and background. Fig. 6(c) is the result modeled under LASSO and solved by ADMM, which is with obvious sparsity, where the ground clutter is suppressed and dominant scatters are enhanced. However, over-sparsity may lead to the loss of structural features. Fig. 6(d) and (e) presents the simulation results formulated under OGL and SGL and solved by ADMM [18], respectively. The recovery result of SGL-ADMM is similar to that of the OGL-ADMM, but neither of them can reach an effective balance between sparsity and structural feature enhancement. The proposed SG-SAR algorithm in Fig. 6(f) gets the best performance in terms of sparsity and structure enhancement. Visually, Fig. 6(f) is closer to the reference image in Fig. 6(a) than others. From the comparison, the proposed SG-SAR algorithm is capable of achieving satisfying features in terms of both sparsity and structure. In terms of quantitative analysis, the correlation value between the ground truth in Fig. 6(a) and (f) is 0.95, which is larger than 0.78 of the SGL-ADMM algorithm in Fig. 6(e), 0.69 of the OGL-ADMM algorithm in Fig. 6(d), and 0.59 of the LASSO-ADMM algorithm in Fig. 6(c). In conclusion, the proposed SG-SAR algorithm is superior to SGL-ADMM, OGL-ADMM, and LASSO-ADMM in terms of the accuracy of feature recovery both qualitatively and quantitatively. Therefore, a balanced performance can be guaranteed, because of the "distributed-cooperative" multitask learning framework applied in this article.

TABLE I
Parameters for Raw "Yak-42" Data

Bandwidth B	400 MHz
Carrier frequency F_c	10 GHz
Range resolution $\rho_r = C/2B$	0.375 m
Pulse repetition frequency(RPF)	200 Hz
Operational Range	10 Km

B. Raw Data Experiments

In this section, the raw radar data of stationary scene and moving target are employed for the verification.

1) Raw Data Experiment of Stationary Scene: First, the raw SAR data was collected by an airborne Ka-band radar in the spotlight mode. The centroid frequency is about 35 GHz, the pulse repetition frequency (PRF) is about 2000 Hz, and the transmitting pulse bandwidth is 600 MHz. The flight height is 3000 m, and the sampling frequency is 800 MHz. The imaging results of LASSO-ADMM, OGL-ADMM, SGL-ADMM, and SG-SAR are shown in Fig. 7. By carefully observing the internal structure of the image, the LASSO-ADMM algorithm effectively enhances the sparse features, but results in the loss of detailed features of the target. Although the OGL-ADMM and SGL-ADMM algorithms in Fig. 7(b) and (c) are better than LASSO-ADMM, the structure recovery in the details is still unsatisfied. Fig. 7(d) is the imaging result of the proposed SG-SAR algorithm, which is distinguished compared with the conventional algorithms in terms of structure recovery. From the comparison, it is obvious that the proposed algorithm is capable of achieving smoothed structures, distinct outlines, and clean background.

2) Raw Data Experiments Moving Target #1: Second, "Yak-42" data are employed for air-target imagery. The radar parameters are given in Table I. The results of the "Yak-42" aircraft by different algorithms are given in Fig. 8. Visually, the structure of the nose is broken in Fig. 8(a), the structure of wings is discontinuous in Fig. 8(b), and the engine in the tail is undistinguished in Fig. 8(c). However, the proposed SG-SAR algorithm shows a preferably continuous structure and preserves the sparsity, simultaneously. A complete structure of

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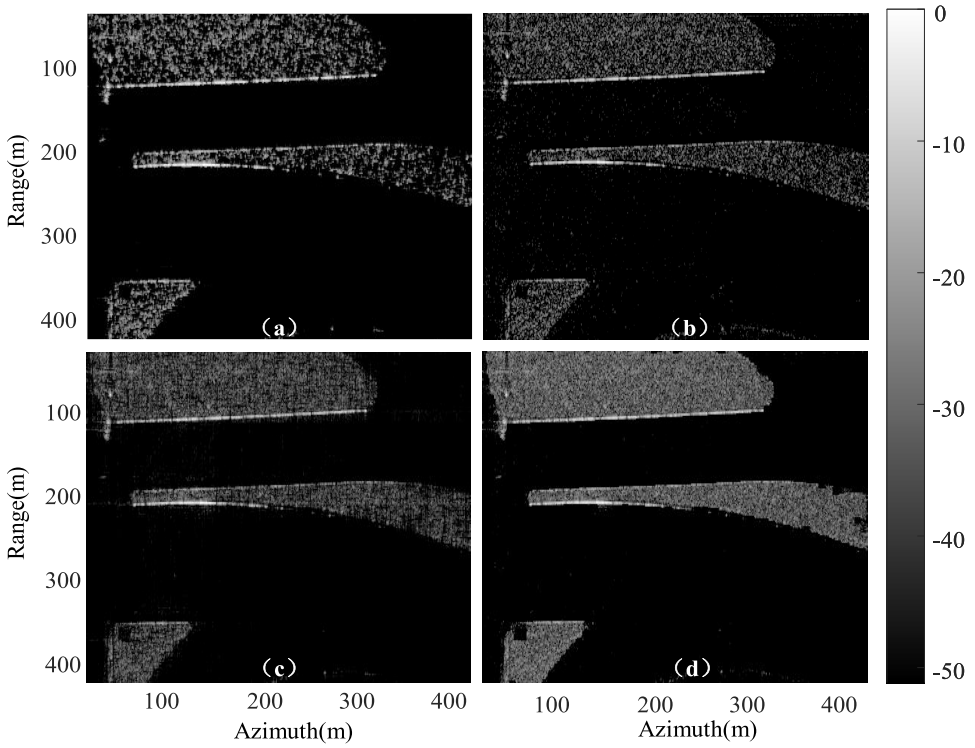


Fig. 7. Raw data experiments of the stationary scene. (a) LASSO-ADMM image. (b) OGL-ADMM image. (c) SGL-ADMM image. (d) SG-SAR image.

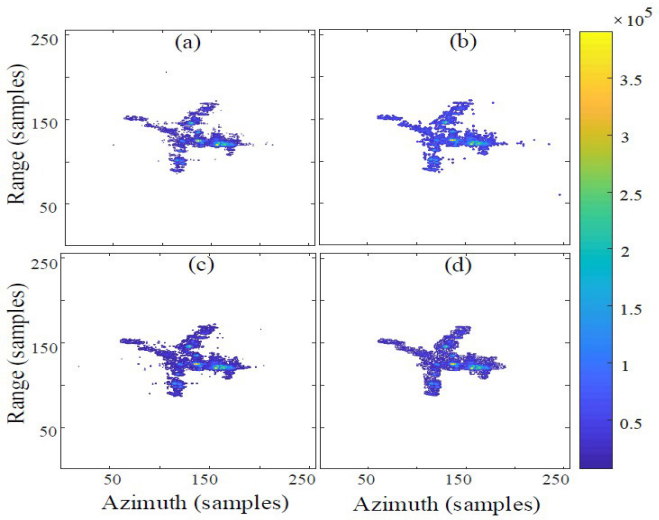


Fig. 8. Experiment of air targets. (a) LASSO-ADMM image. (b) OGL-ADMM image. (c) SGL-ADMM image. (d) SG-SAR image.

the air target is guaranteed during the recovery, and simultaneously, a high SNR target image is obtained.

3) Raw Data Experiments Moving Target #2: Next, the airborne Gotcha dataset [42] is applied for imagery of ground moving target. The canonical Gotcha data was collected by a multichannel (three channels) X-band (9.6 GHz) radar of the Air Force Research Laboratory (AFRL). Details about the dataset [42] include the chirp bandwidth is about 640 MHz, the PRF is about 2172 Hz, and the spatial resolution can be up to 0.3 m. A cooperative ground car, named Durango target, was illuminated within the radar beam. Fig. 9(a) is

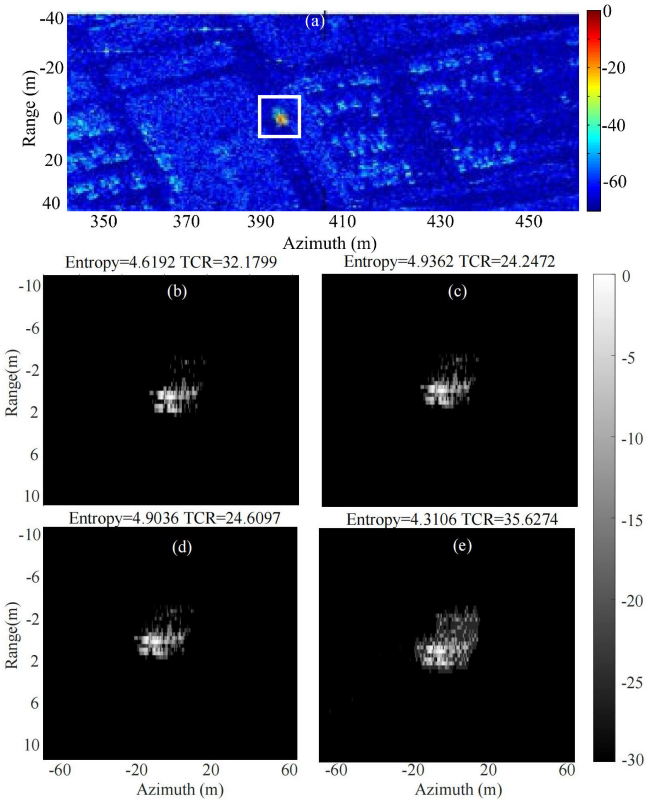


Fig. 9. Raw data experiment for ground moving target. (a) Relocated and refocused Durango target. (b) LASSO-ADMM image. (c) OGL-ADMM image. (d) SGL-ADMM image. (e) Proposed SG-SAR image.

the refocused and relocated Durango target on the stationary ground scene. Fig. 9(b)–(e) are enlarged image from Fig. 9(a)

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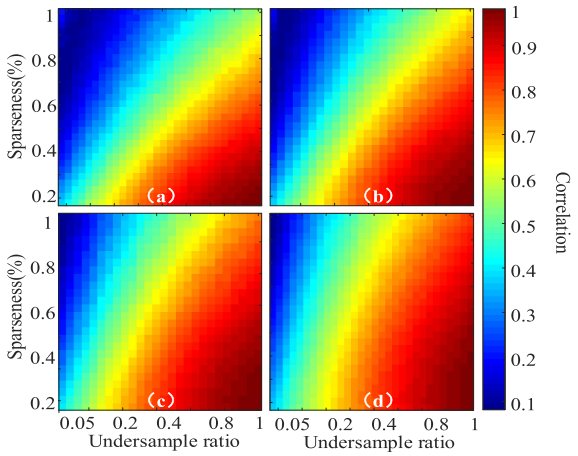


Fig. 10. PTD of SR and USR under SNR equals −5 dB. (a) LASSO-ADMM. (b) OGL-ADMM. (c) SGL-ADMM. (d) Proposed SG-SAR.

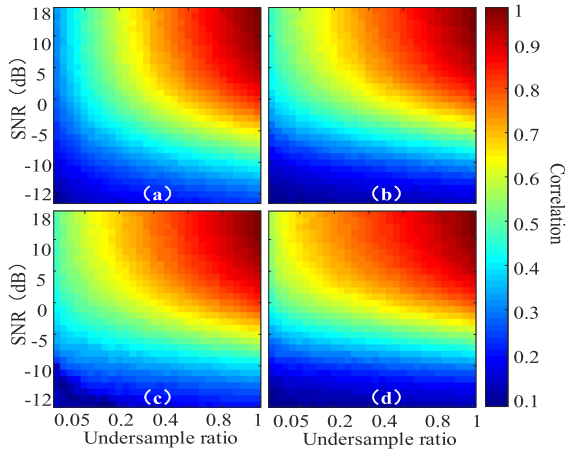


Fig. 11. PTD of SNR and USR under SR equals 50%. (a) LASSO-ADMM. (b) OGL-ADMM. (c) SGL-ADMM. (d) Proposed SG-SAR.

where the ground clutter is suppressed. In terms of visual perception, it can be noted that the target in Fig. 9(e) has a relatively complete profile and relatively rich details. In terms of quantitative indicators, image entropy and target-to-clutter ratio (TCR) are provided to show the superior performance of the proposed algorithm. The smaller the image entropy, the higher the focusing quality of the target image. The larger the TCR, the better the image contrast. Therefore, the proposed algorithm performs better than the conventional SGL-ADMM, OGL-ADMM, and LASSO-ADMM both qualitatively and quantitatively.

C. Performance Analysis

To further evaluate the performance of the proposed algorithm quantitatively, the phase transition analysis is employed. The concept of phase transition was first proposed in [43], which represents the variation trend of algorithm performance under different conditions. The larger the area with a high correlation value, the better performance of the feature recovery or enhancement. In this article, an extended version of the phase transition diagram (PTD) is used. An increased number of arguments are applied during performance validation. PTD is performed in a Monte Carlo (MC) manner. By repeating the experiments independently, the mean of the correlation between the experimental and reference (ground truth) images expresses the accuracy of the algorithm under testing. By fulfilling the tests in various conditions, the area with high correlations denotes a high probability of robustness and accuracy. The PTD validation in this section is based on the simulated data in Fig. 6. By setting the SNR of the polymorphic simulated data to be −5 dB, the PTD of the sparseness ratio (SR) and USR is shown in Fig. 10. It can be noted that the boundaries of different correlation values form the phase transition curves, which are basically similar to each other. The proposed SG-SAR algorithm is obviously superior to the others in terms of the size of the highly correlated area.

By setting the simulated SR to be 50%, the PTD of the SNR and USR is shown in Fig. 11. Obviously, the proposed SG-SAR algorithm is with a larger area than others. The larger the area, the higher the accuracy for the feature recovery. Therefore, the SG-SAR is more accurate and robust than the conventional algorithms, especially in low SNR, SR, and USR circumstances.

VII. Conclusion

In this article, a novel SG-SAR imaging algorithm is proposed to enhance the structural continuity and sparsity of the target of interests, simultaneously and cooperatively. It is developed under the generalized ADMM framework that guarantees superior robustness and balanced accuracy and efficiency. Specifically, the proposed SG-SAR algorithm incorporates the spatially variant morphological norm for the clustering feature of the scatterers of the targets of interest, and the proximal mapping corresponding to the morphological norm is derived. Besides, a closed-form solution for the sparse ℓ_1 regularizer is derived as a complex-valued version of the soft-thresholding operator. Furthermore, the structural and sparse features are incorporated within the ADMM framework in a multitask learning manner. The primal variable is split into two feature variables, and both of them are optimized as distributed learning. Two additional dual-variables are introduced to evaluate the accuracy of the feature recovery. Finally, the multifeature variables and dual-variables are combined and coordinated to update the primal image. Due to the spatially variant MR and the "distributed-cooperative" learning scheme, superior performance in terms of structural continuity enhancement can be obtained. Both simulated and various raw data experiments validate the proposed algorithm. Comparisons with conventional algorithms show the superiority of the proposed algorithm.

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